

Milind Bansal

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SUMMARY

Computer Science and AI student building ML pipelines and LLM-backed products end to end: candidate retrieval, gradient-boosted models, leakage-safe validation, and deployed TypeScript/Python applications. Seeking a year-long AI engineering internship.

EDUCATION

Newton School of Technology at Rishihood University

2024-2028

B.Tech in Computer Science and Artificial Intelligence

CGPA: 9.28/10

Minor in Finance

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

ML & Data: scikit-learn, LightGBM, XGBoost, pandas, NumPy, TF-IDF, SHAP, cross-validation, entity resolution

AI & Backend: OpenAI SDK, structured outputs, FastAPI, Pydantic, Zod, PostgreSQL, Prisma, Redis, BullMQ

Web & Tools: React, Next.js, Express, REST APIs, Git/GitHub, unittest, pytest

PROJECTS

Amazon ML Challenge 2026 — Business Entity Resolution

Team competition entry (Team Noobs) · Co-built the matching pipeline

- Matched **1.73M** test businesses across three sources with multi-route inverted-index blocking (character TF-IDF n-grams, rare tokens, postal and house-number keys) feeding a LightGBM pair classifier.
- Raised held-out blocking recall from **0.843 to 0.907** and held-out macro F0.5 from **0.848 to 0.893** through wider candidate caps and threshold tuning on unseen entities.
- Diagnosed over-prediction in a test country absent from training (8.6 vs 3.5 links per record) and added a data-only filter, lifting public-leaderboard F0.5 from **0.819 to 0.870**; covered by 28 unit tests.

C-MAPSS — Turbofan Remaining-Useful-Life Prediction

Four-person team · 4 merged pull requests · [GitHub](#)

- Implemented seeded engine-level splits and fold-fitted operating-regime normalization so no test-engine statistics leak into training; added regression tests for invalid and unfitted use.
- Built matched cross-validated comparisons of raw-sensor and engineered-feature Random Forest, XGBoost and Elastic Net models; engineered Random Forest cut FD001 holdout RMSE from **30.8 to 26.5** cycles.
- Reported that mean CV RMSE did not confirm a general feature advantage, and added tree SHAP analysis of the sensors driving predictions.

FeedbackOS — LLM-Assisted Feedback Triage

Solo project · [GitHub](#) · [Live](#)

- Built a deployed Next.js/TypeScript product that classifies feedback theme, sentiment, and product area with OpenAI structured output validated by Zod, routing low-confidence results to human review.
- Decoupled ingestion from classification with Redis/BullMQ workers and bounded concurrency; added capped CSV ingestion and organization-scoped cached summaries.

TruthLens — NLP Classification & Model Evaluation

Solo project · [GitHub](#) · [Live](#)

- Built a TF-IDF and logistic-regression misinformation classifier with a FastAPI service, SHAP token explanations, drift checks, and fail-closed serving.
- Enforced a **0.75** F1 release gate: held-out LIAR F1 of **0.5648** failed it, so the model was blocked from release instead of shipped.